



**Innovation & Entrepreneurship Hub for Educated Rural Youth**  
(SURE Trust - IERY)

## **TECHNICAL PROJECT REPORT**

### **PROJECT TITLE**

**Noise-Free Audio & Voice Recorder PCB (ESP32 + I2S)**

### **Domain of the Project**

PCB Designing & Embedded Audio Hardware

### **Team Mentor**

Mr. Paramesh  
(*Mentor – SURE Trust*)

### **Submitted By**

**Mr. L. Bharath Kumar**

### **Project Duration**

1 Month

# DECLARATION

The project titled “**Noise-Free Audio & Voice Recorder PCB (ESP32 + I2S)**” has been successfully mentored by Mr. Paramesh and organized by **SURE Trust**, with the primary objective of empowering educated rural youth through hands-on industrial project training. This practical exposure bridges academic concepts with real-world employer requirements.

I hereby declare that the work presented in this technical report was carried out by the team member listed below, reflecting genuine effort, high standards of hardware design, and comprehensive technical learning.

## Team Member Details:

- **Mr. L. Bharath Kumar**

Signature: \_\_\_\_\_

**Mentor's Signature:** \_\_\_\_\_

**Name:** Mr. Paramesh

**Designation:** Mentor – SURE Trust

**Executive Director:**

**Prof. Radhakumari**

SURE Trust

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# 1 Executive Summary

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The primary objective of this project was to design and fabricate a professional, ultra-low-noise embedded hardware development board centered around the ESP32-WROOM-32 microcontroller dedicated to high-fidelity audio capture and voice recording. To achieve crystal-clear, noise-free acoustic performance, the board integrates dual ICS-43434 I2S digital MEMS microphones directly coupled with a high-efficiency TPS54331D step-down DC-DC switching regulator and rigorous power filtering stages. Furthermore, the system incorporates a DS3232M high-precision Real-Time Clock (RTC) with coin cell battery backup for precise timestamping of recordings, alongside a MicroSD card socket (ST-TF-003A) operating via SPI for extensive local data logging.

Both the complete schematic capture and the professional multi-layer PCB layout were meticulously engineered using KiCad software, adhering strictly to high-speed digital audio design rules, differential routing practices, clean ground plane partitioning, and thorough Design Rule Check (DRC) verification. Ultimately, the project successfully consolidated complex audio acquisition, precise timing, and local storage into a compact, reliable, and production-ready hardware platform.

## 2 Introduction

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Modern digital audio systems and voice recorders require exceptionally low noise floors, stable power supplies, and high-speed synchronous data interfaces (such as I2S) to capture acoustic signals without distortion or electromagnetic interference. Traditional audio recording modules often suffer from ground noise, power ripple interference, and clock jitter. To overcome these limitations and gain direct industrial hardware design experience, this project focuses on engineering a specialized, noise-free audio and voice recorder printed circuit board from the ground up.

Optimizing power distribution and signal routing is critical in mixed-signal audio hardware to prevent high-frequency digital noise from corrupting sensitive analog and digital microphone outputs. Realizing practical audio hardware bridges theoretical microprocessor interfacing with actual consumer electronics manufacturing standards.

### 2.1 Problem Statement and Goals of the Project

Recording high-quality audio using generic microcontroller development boards typically results in audible background hum and RF interference caused by unshielded traces and shared noisy power rails. Designing and wiring discrete audio breakout modules also increases wiring complexity and system vulnerability. The primary goal of this project is to eliminate these issues by integrating an ESP32 microcontroller, dual I2S digital

microphones, an accurate RTC, and local MicroSD storage onto a single, optimized, low-noise printed circuit board (PCB).

Isolating digital switching supplies from sensitive microphone VDD lines minimizes parasitic noise injection. Consolidating core features provides a robust, standalone hardware platform tailored for professional voice recording and acoustic data logging.

## 2.2 Scope and Limitations of the Project

The scope of this project encompasses complete schematic capture, component selection, multi-layer PCB layout routing, power supply design, audio peripheral integration, storage interface routing, and the preparation of manufacturing outputs including Gerber files and Bills of Materials (BOM). However, firmware audio processing algorithms and acoustic testing after physical fabrication remain outside the scope of this hardware-centric project.

The layout scope covers stackup design, ground plane separation rules, and component footprint verification. Post-fabrication hardware bring-up and embedded firmware development are explicitly excluded from the current hardware deliverables.

## 2.3 Innovation Component in the Project

Unlike standard evaluation kits, this custom design integrates multiple specialized industrial subsystems onto a single board: an ESP32-WROOM-32 module, dual ICS-43434 I2S digital MEMS microphones for stereo/multi-channel acoustic capture, a DS3232M I2C real-time clock with battery backup for autonomous timestamping, a MicroSD card interface, and a high-current TPS54331D step-down regulator. This comprehensive integration makes the board ideal for advanced field voice recording and sound monitoring applications.

The inclusion of dual I2S microphones enables advanced beamforming and stereo capture capabilities. The dedicated RTC ensures that every recorded audio file is accurately cataloged with precise date and time metadata.

# 3 Project Objectives and Methodology

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## 3.1 Clearly Defined Objectives and Goals of the Project

The main objectives of this project are to design a complete, low-noise audio recorder development board based on the ESP32 microcontroller and to capture the complete circuit schematic using KiCad. The goals include designing a professional multi-layer PCB following industry-standard design practices for mixed-signal audio, integrating dual I2S digital microphones (ICS-43434), a DS3232M RTC, a MicroSD card socket, and

a TPS54331D power regulation circuit. Additionally, the project aims to ensure zero design errors through Electrical Rule Checks (ERC) and Design Rule Checks (DRC), generate manufacturing outputs like Gerber files and Bills of Materials (BOM), and enhance practical competence in embedded hardware development.

### 3.2 Expected Outcomes and Deliverables

The expected outcomes of this project include an error-free schematic capture and a professional PCB layout designed in KiCad. The key deliverables consist of manufacturing-ready Gerber and BOM files, a compact audio recording platform, and significantly improved technical proficiency in high-speed digital and audio PCB design.

### 3.3 Methods and Tools Used

- **Methods/Technology used:** The project followed a systematic engineering methodology tailored for audio hardware, encompassing requirement analysis, component selection, schematic design with electrical checks, PCB layout routing, ground plane optimization, design rule verification, and manufacturing output generation.
- **Tools/Software used:** The project was designed using KiCad EDA, serving as the primary environment for schematic capture, PCB routing, design verification (ERC/DRC), 3D visualization, and manufacturing file generation.
- **Data collection approach:** Technical parameters, acoustic specifications, pin mappings, and electrical layout requirements were gathered from official component datasheets and application notes to guarantee hardware reliability and acoustic fidelity.

## 4 Project Architecture and Working Screenshots Explanation

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### 4.1 Project Architecture & Flow Diagrams

The system follows a structured modular architecture optimized for low-noise audio acquisition. Below are three different visual representations showcasing the power distribution, core microcontroller processing, and peripheral subsystem integration.

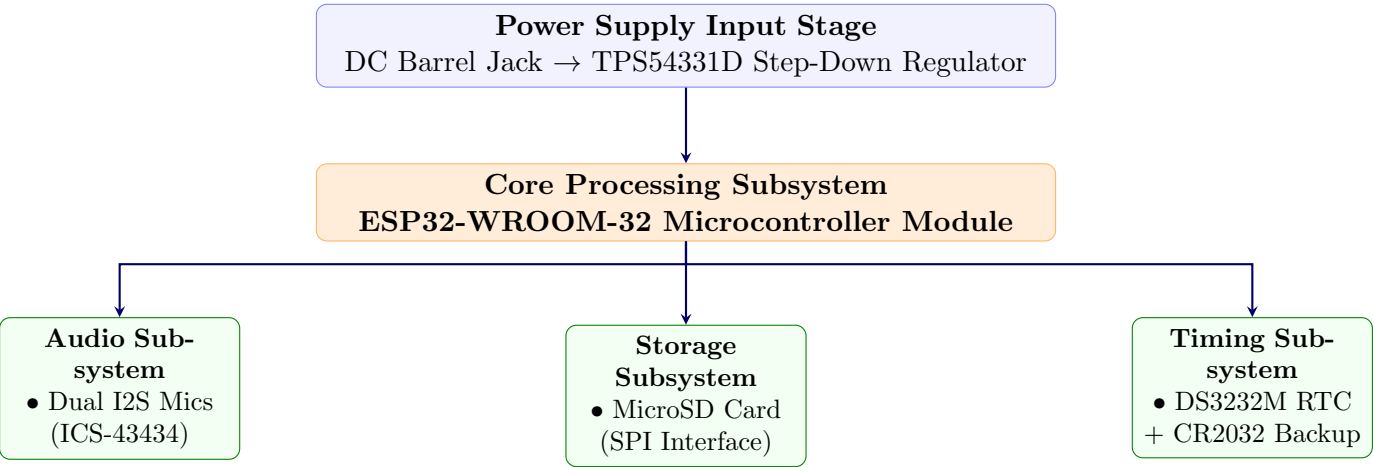


Figure 1: System Architecture Flowchart - Style 1 (Graphical Blocks)

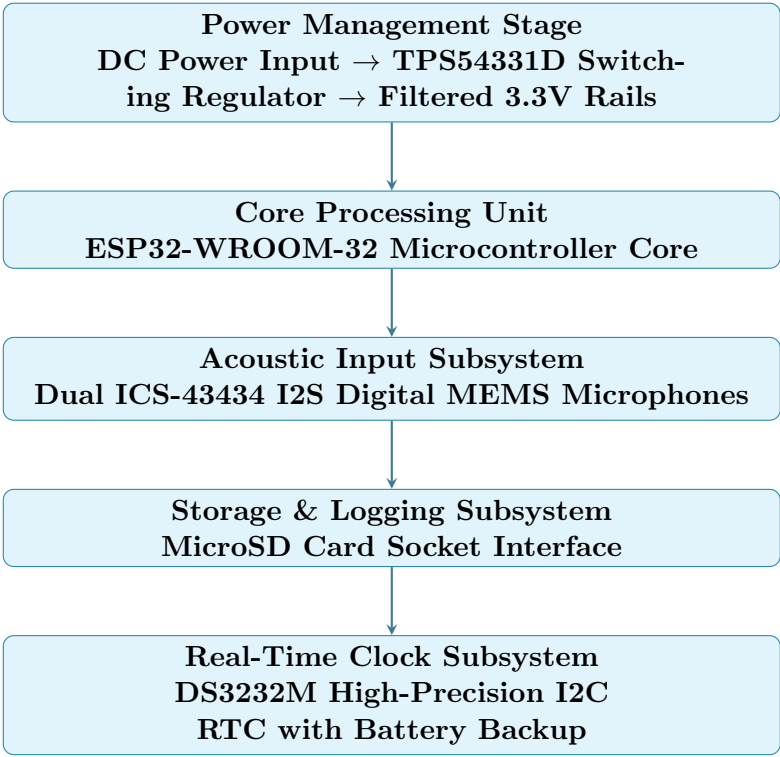


Figure 2: System Architecture Flowchart - Style 2 (Structured Box Layout)

#### 4.1.1 Flowchart Style 1: Graphical Block Architecture (TikZ)

#### 4.1.2 Flowchart Style 2: Structured Box Architecture (TikZ Layout)

#### 4.1.3 Flowchart Style 3: Sequential Linear Pipeline Architecture

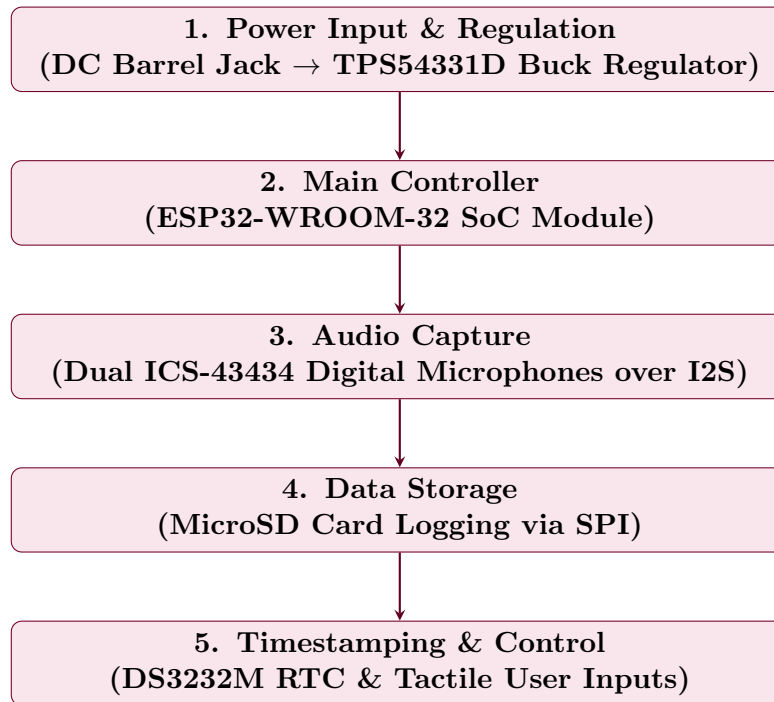


Figure 3: System Architecture Flowchart - Style 3 (Sequential Pipeline)

- **Power Flow & Filtering:** External power enters via the DC Barrel Jack and is regulated by the TPS54331D switching regulator, delivering clean operating voltages across the board.
- **Core Processing Unit:** The ESP32-WROOM-32 coordinates audio data streams, manages file storage, and handles wireless connectivity.
- **Peripherals & Subsystems:** Integrates dual digital I2S microphones for clean audio capture, an RTC for accurate file timing, and a MicroSD card socket for large-scale audio storage.

## 4.2 Final Project Working Screenshots & Visual Verification

### 4.2.1 Complete Schematic Diagram

The schematic captures all functional blocks—power regulation, ESP32 microcontroller core, dual I2S microphone circuits, DS3232M RTC, and MicroSD card interface—verified through Electrical Rule Checks (ERC).

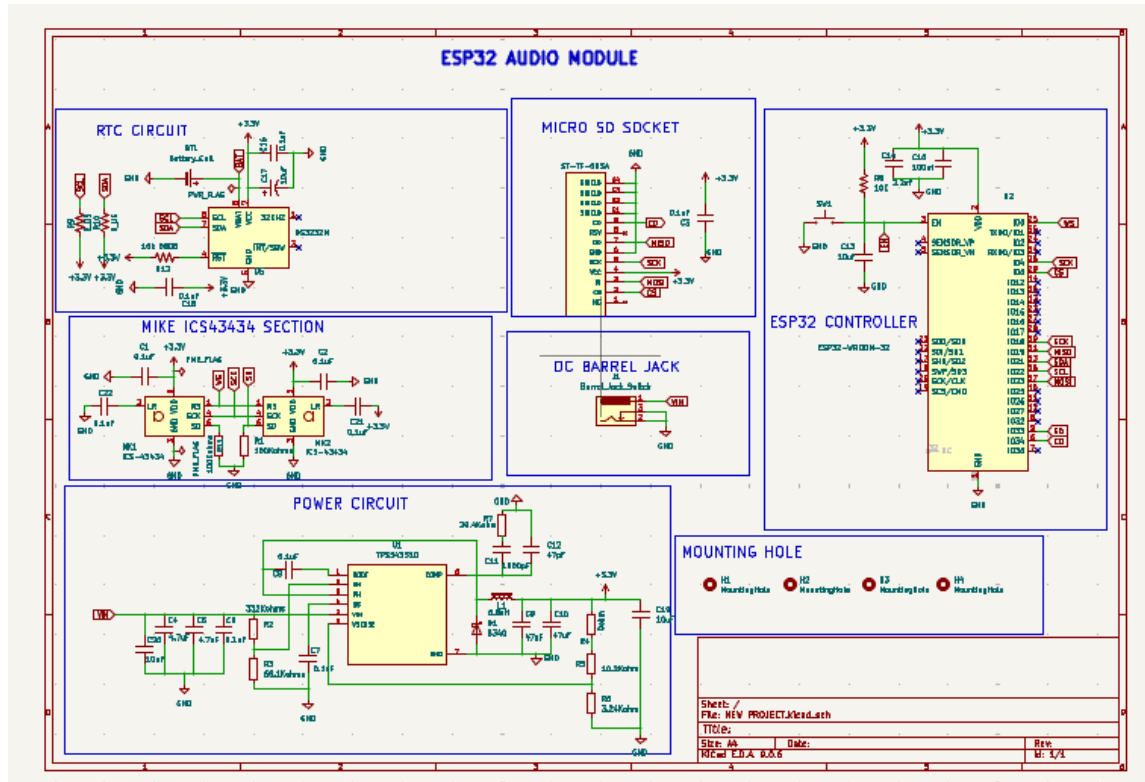


Figure 4: Complete Circuit Schematic of the Noise-Free Audio Recorder (ESP32 + I2S)

#### 4.2.2 PCB Layout

The multi-layer board view illustrates the precise layout implementation, showing component placement, high-speed signal routing on external copper layers, and dedicated ground plane continuity.

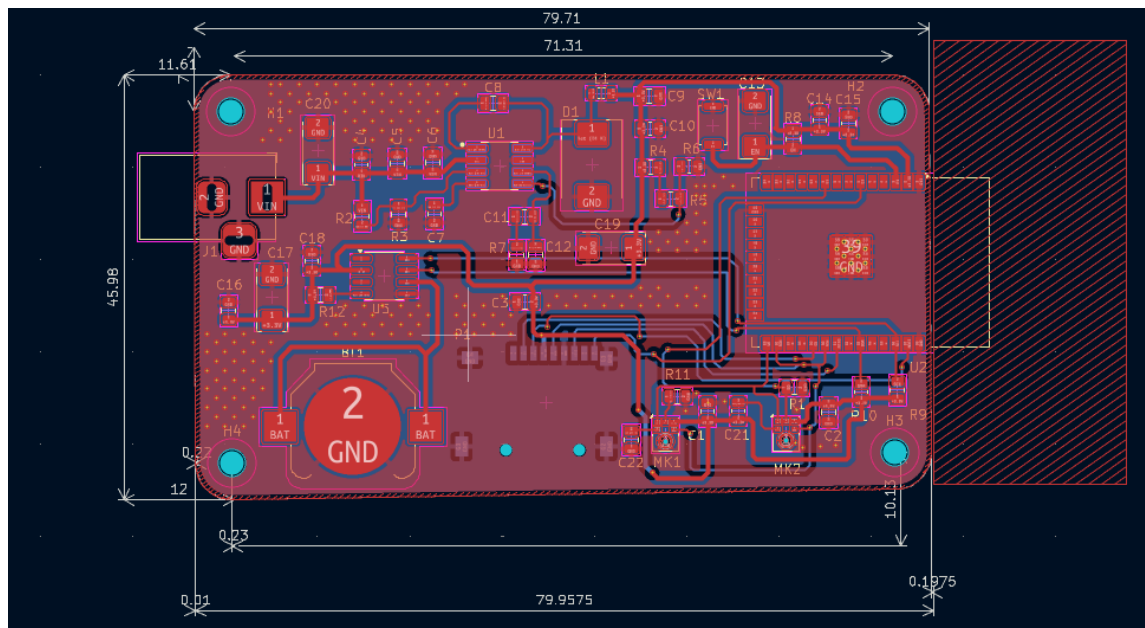


Figure 5: PCB Layout - Full Board Overview with Dimensions and Routing

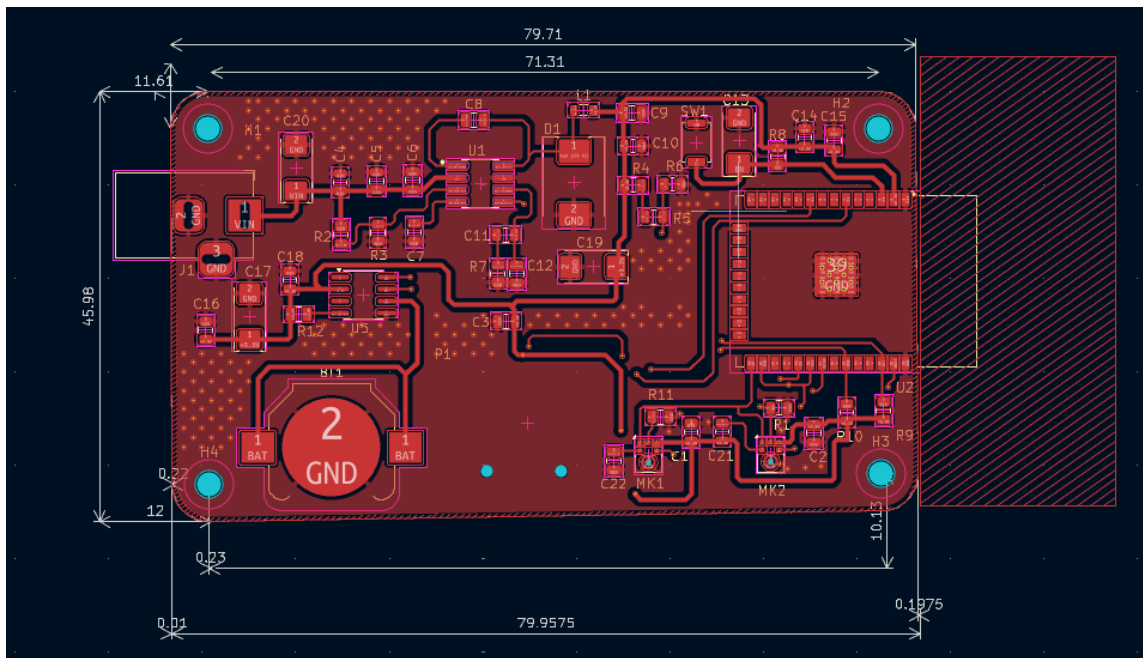


Figure 6: PCB Layout - Top Copper Routing and Component Placement

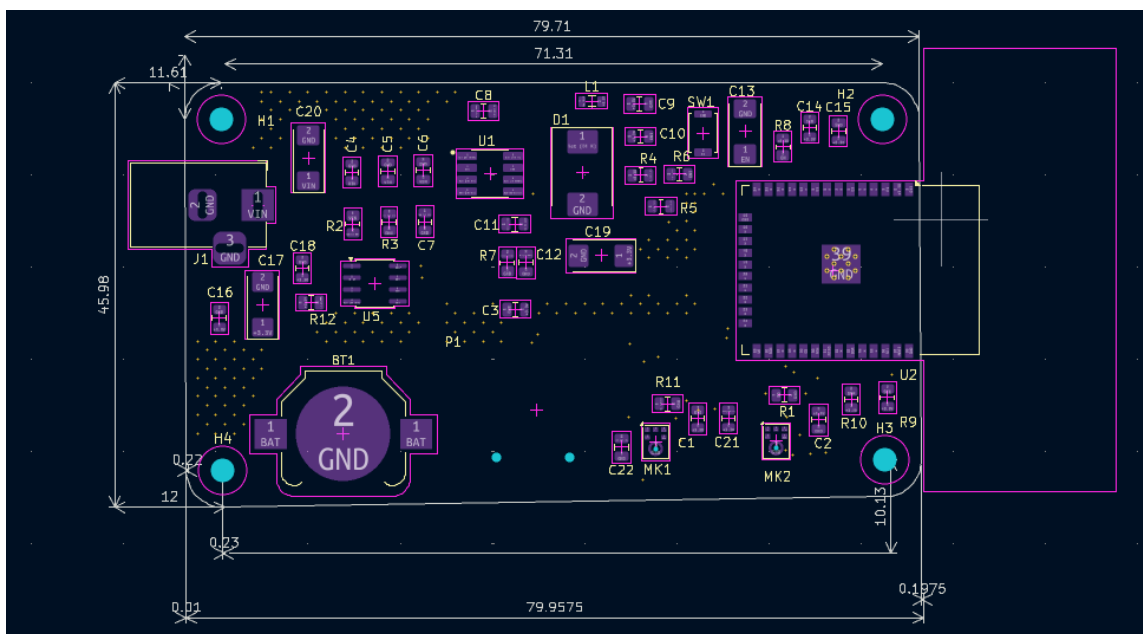


Figure 7: PCB Layout - Silkscreen and Component Reference Designators

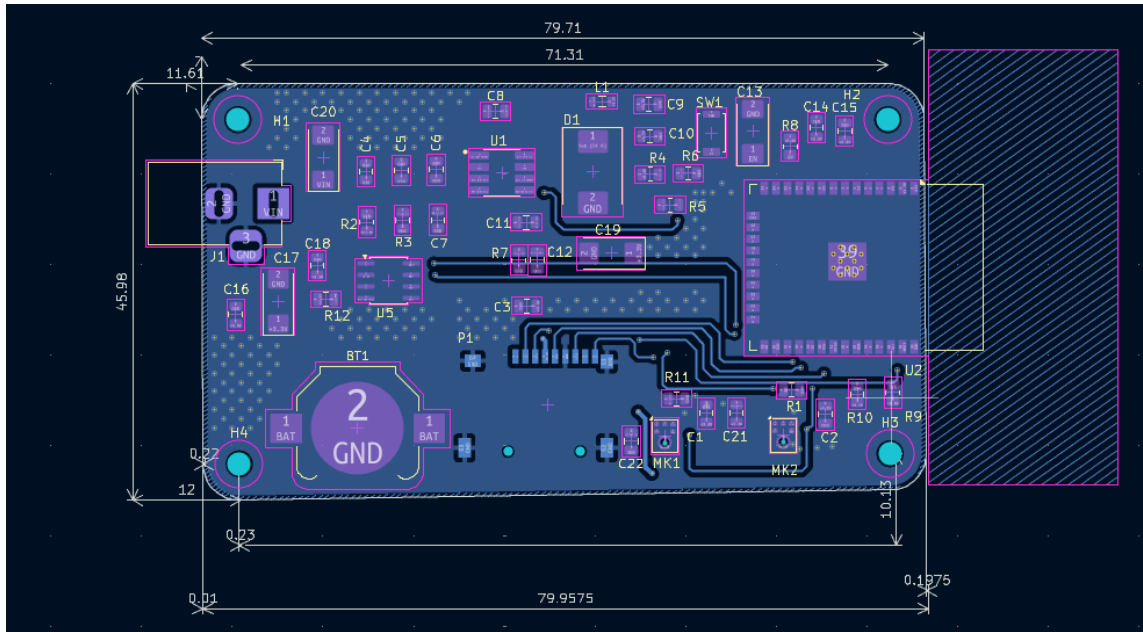


Figure 8: PCB Layout - Inner and Trace Routing Inspection

#### 4.2.3 3D View of the PCB

The realistic 3D rendered view provides physical verification of the assembled board, confirming proper component heights, correct connector orientations, and mechanical clearances.

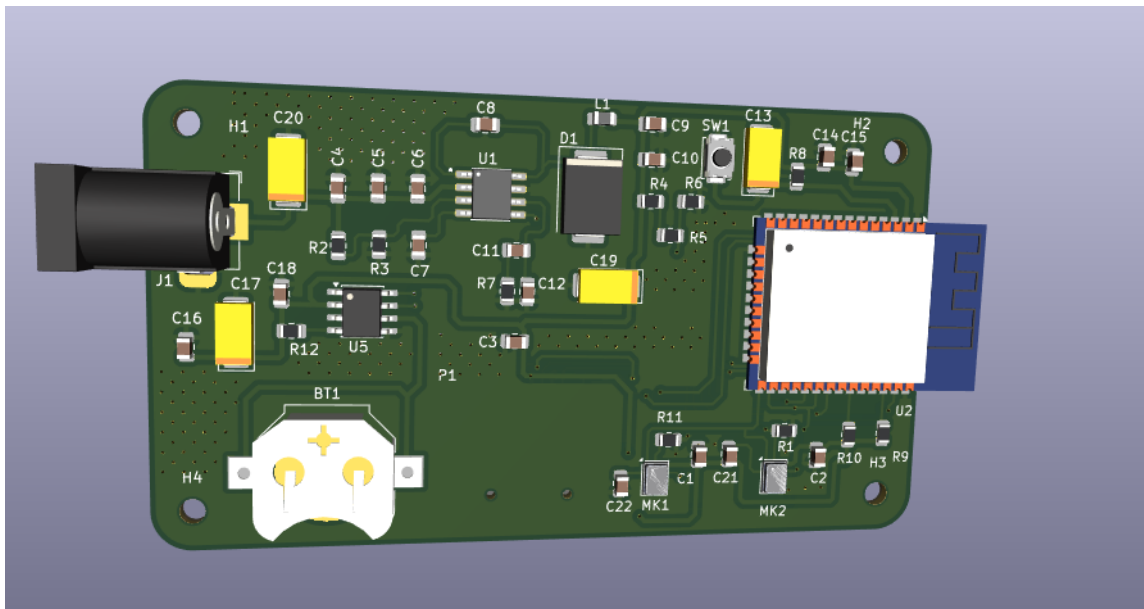


Figure 9: 3D Rendered View of the PCB - Top Side Assembly (Including DC Jack and Battery Holder)

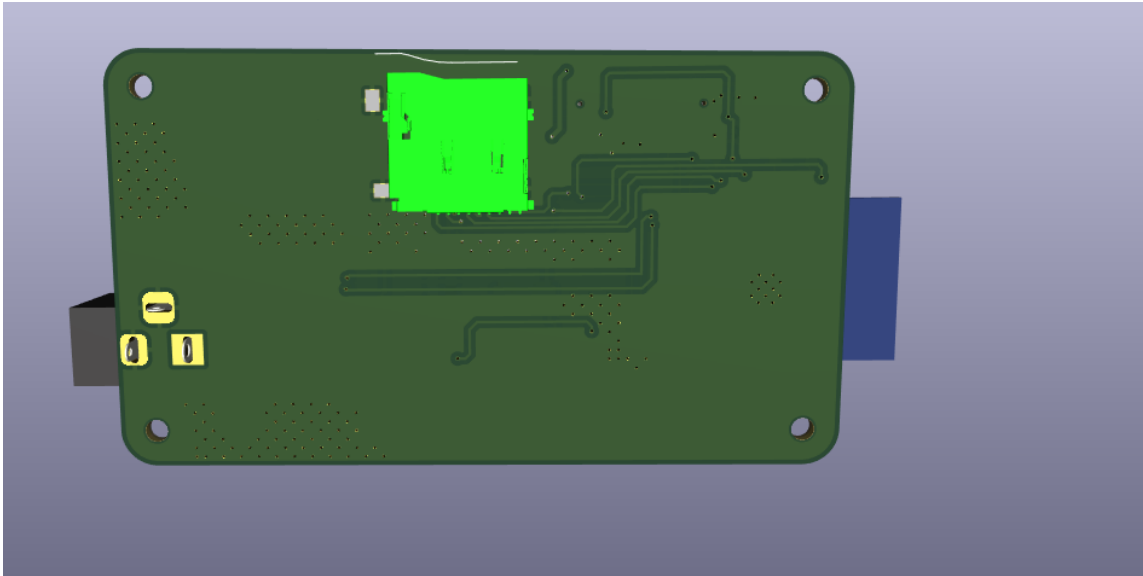


Figure 10: 3D Rendered View of the PCB - Bottom Side Assembly (Highlighting MicroSD Card Socket)

## 5 Learning and Reflection

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### 5.1 Document Every Team Member's New Learnings

- **Low-Noise Audio Design Principles:** Acquired practical knowledge in designing noise-free power supply topologies and layout isolation techniques specifically for sensitive I2S digital MEMS microphones.
- **Multi-Peripheral Hardware Integration:** Mastered interfacing high-speed storage (MicroSD over SPI), precision real-time clocks (DS3232M over I2C), and wireless microcontrollers (ESP32).
- **Switching Regulator Design:** Gained hands-on proficiency in implementing the TPS54331D step-down regulator circuit, managing component placement, feedback loops, and thermal dissipation.
- **Design Validation & Output Generation:** Executed rigorous ERC and DRC checks, successfully generating manufacturing-ready Gerber files, drill files, and BOM outputs.

### 5.2 Document Every Team Member's Experience

- **End-to-End Hardware Journey:** Taking a noise-free audio recorder project from initial schematic design to a professional multi-layer PCB layout provided invaluable hands-on engineering experience.

- **Overcoming Technical Hurdles:** Addressed challenges in high-frequency power switching noise reduction, microphone placement constraints, and compact routing through careful datasheet analysis.
- **Professional Growth Confidence:** Substantially enhanced technical confidence, problem-solving skills, and a comprehensive understanding of modern embedded audio hardware workflows.

## 6 Conclusion and Future Scope

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### 6.1 Recap Objectives and Achievements

The primary objective of designing a professional, noise-free audio and voice recorder development board based on the ESP32 microcontroller was successfully achieved. The project milestones were fully realized through meticulous schematic capture and robust PCB layout design in KiCad. Essential subsystems—including DC power input, TPS54331D regulation, dual ICS-43434 I2S microphones, DS3232M RTC, and MicroSD card storage—were seamlessly integrated onto a compact platform. Rigorous verification through ERC and DRC ensured that the hardware architecture is entirely error-free and production-ready.

### 6.2 Future Scope of this Project

- **Embedded Firmware Development:** Developing optimized firmware to stream and save uncompressed audio files directly to the MicroSD card.
- **Wireless Streaming:** Utilizing the ESP32's Wi-Fi and Bluetooth capabilities to stream recorded audio live to web servers or cloud dashboards.
- **Voice Activation (VOX):** Implementing voice activity detection to record only when acoustic events occur, optimizing power and storage.
- **Advanced Filtering & DSP:** Adding digital filtering stages to further enhance voice clarity in noisy environments.
- **Battery Integration:** Adding a Li-ion charging and battery management circuit for portable field operations.

## 7 References

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1. Espressif Systems, *ESP32-WROOM-32 Datasheet*, Wi-Fi and Bluetooth Module Datasheet.
2. TDK InvenSense, *ICS-43434 Datasheet*, Low-Power Multi-Mode Digital InvenSense MEMS Microphone.
3. Texas Instruments, *TPS54331D 3-A, 28-V Input, Step-Down Voltage Regulator Datasheet*.
4. Analog Devices / Maxim Integrated, *DS3232M  $\pm 5\text{ppm}$ , I2C Real-Time Clock with SRAM Datasheet*.
5. KiCad Automated Design Association, *KiCad EDA Reference Manual and Documentation*.